Calibration Basics

Pre-Work Module 05

ECBS5200 — Practical Deep Learning Engineering

What does "90% confident" mean?

Suppose your model predicts **90% confidence** on 100 different complaints.

If the model is **well-calibrated**, about **90 of those 100** should actually be correct.

That's it. That's calibration.

- **Calibrated**: predicted probability matches observed frequency
- If the model says 80%, it should be right ~80% of the time
- If the model says 50%, it should be right ~50% of the time

The overconfidence problem

Most neural networks are **overconfident**.

They routinely say **95% confident** when they're actually right only **70% of the time**.

Model says...	Actually correct...
95%	~70%
80%	~55%
60%	~45%

This isn't a bug in one model — it's a **systematic property** of how neural networks are trained.

Modern deep networks tend to produce overly sharp probability distributions.

Why calibration matters in production

Imagine you deploy a complaint classifier with a business rule:

> "Only auto-route complaints where the model is **>80% confident**"

If the model is **well-calibrated**, this threshold is meaningful.
Complaints above 80% really are almost certainly correct.

If the model is **overconfident**, your threshold is **meaningless**.
The model says 80% for predictions it gets wrong 40% of the time.

Miscalibration turns every probability threshold into a lie.

Reliability diagrams

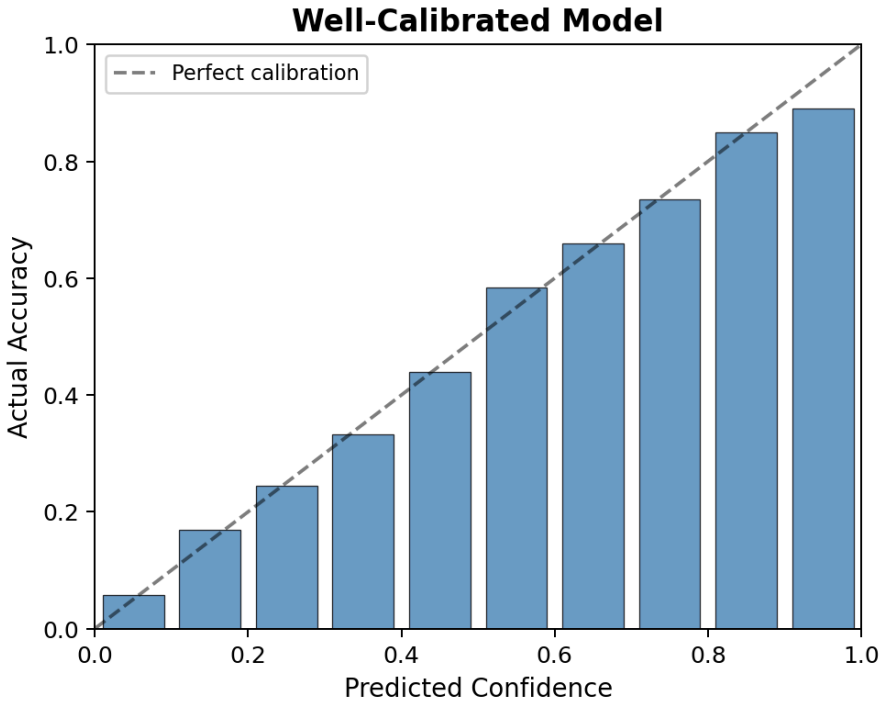
The standard tool for visualizing calibration is the **reliability diagram**.

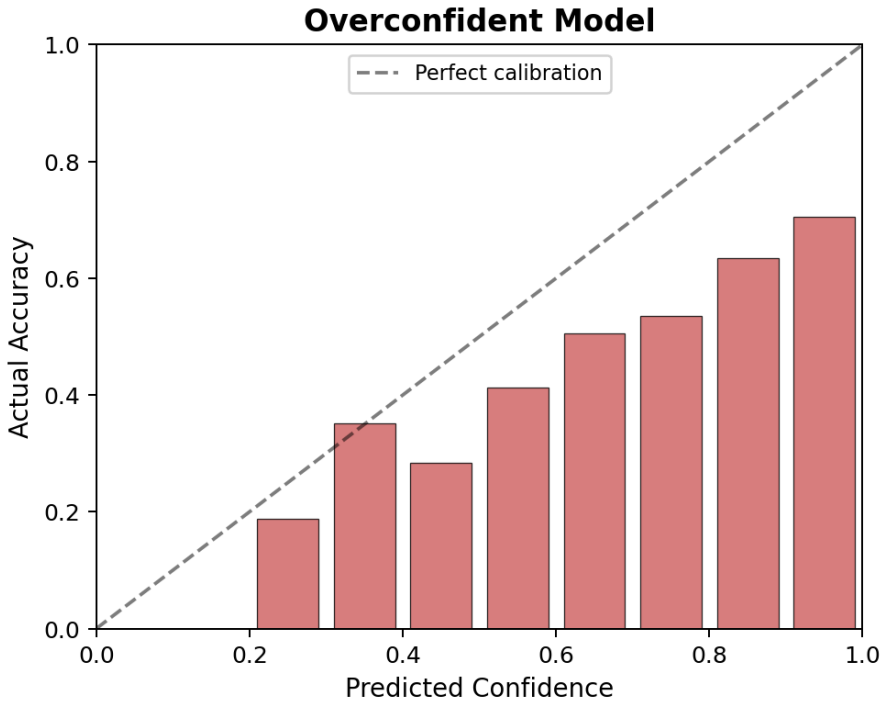
- **X-axis**: predicted confidence (binned, e.g., 0.0–0.1, 0.1–0.2, ...)
- **Y-axis**: actual accuracy within that bin
- **Perfect calibration** = the diagonal line

A curve **below the diagonal** at high confidence = overconfidence.

Reliability diagrams: reading them

Pattern	Meaning
Points on the diagonal	Well-calibrated
Points below diagonal (high conf.)	Overconfident
Points above diagonal (low conf.)	Underconfident
Histogram shows most predictions at extremes	Model is very "decisive"





What we won't do yet

We are **not** going to fix calibration in this module.

Techniques like **temperature scaling** and **Platt scaling** exist to recalibrate models after training. We'll cover those in **Week 4** of the course.

For now, the goal is:

1. **Know** that model probabilities are often not trustworthy
2. **Understand** why this matters for any system that uses probability thresholds
3. **Recognize** a reliability diagram when you see one

That's sufficient for pre-work. The fix comes later.

Key takeaways

1. **Calibration** = predicted probabilities match observed frequencies
2. Most neural networks are **systematically overconfident**
3. Overconfidence makes **probability thresholds unreliable** in production
4. **Reliability diagrams** are the standard visualization tool
5. We'll learn to **fix** calibration in Week 4 — for now, just know it matters

Next up: Module 06 — LoRA & PEFT Basics